

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

20ME504 / 17ME504 - METROLOGY AND MEASUREMENTS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Missing data if any may suitably be assumed and specified.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain meter with regards to Wavelength standards and light standards. Why is material standards not preferred?	10	L*3	1	2
	b) Explain with a sketch the principle of laser interferometer.	10	L2	2	2
2.	a) Define comparator. Explain the optical type of comparator with a sketch and state its needs.	10	L2	2	1
	b) Illustrate and discuss the working principle of LVDT as an electrical comparator and write the advantages and disadvantages of the same.	10	L2	1	2
3.	a) Explain how the gear tooth Vernier caliper is used to determine various gear parameters.	10	L2	2	2
	b) Explain the wringing phenomenon of slip gauges. Also build the dimension 210.6825 using M87 and M112 set of slip gauges.	10	L2	1	2
Unit – II					
4.	a) Design a general purpose 'GO' and 'NO-GO' plug gauge for checking machined hole of size 22 D8. What should be the dimensions of the gauge, if unilateral system of tolerances is incorporated? Assume: 22 mm lies in the diameter step of 18 - 30 mm and fundamental deviation for hole = 16 D0.44.	10	L3	3	2
	b) Differentiate between accuracy and precision with proper illustrative examples.	10	L1	4	1
5.	a) With a neat sketch and suitable examples, explain the generalized measurement system.	10	L2	4	2
	b) Illustrate with sketch the different types of fits.	10	L2	3	1
6.	a) Explain the constructional features and working principle of X-Y plotter.	5	L2	4	2
	b) Explain the principle of Electrical transducer, explain Wheatstone network.	5	L2	4	2
	c) Explain the concept of interchangeability and selective assembly with illustrative examples.	10	L2	3	2
Unit – III					
7.	a) Describe the working principle of proving ring in the material testing laboratory for the measurement of compressive force.	8	L2	5	2
	b) What are the various thermocouple materials used for wide ranges of temperature measurements?	6	L2	5	2
	c) Discuss briefly the working principle of resistance thermometers.	6	L2	5	1
8.	a) With a neat sketch explain the device used for measurement of temperature of furnaces used in steam power plants.	8	L2	5	1
	b) With a neat diagram explain the Prony brake dynamometer used for torque measurement.	6	L2	5	2
	c) What are the precautions to be taken while mounting the strain gauges?	6	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
